

DO NOT REMOVE FROM EXAM VENUE

University of Plymouth

MODULE CODE: ELEC345

TITLE OF PAPER: High Speed Communications

TIME ALLOWED

3 Hours

DATE

Tuesday 14th January 2025

TIME

09:00 – 12:00

FACULTY

Science and Engineering

SCHOOL

Engineering, Computing and Mathematics

ACADEMIC YEAR

2024/25

STAGE

Three

INSTRUCTIONS TO CANDIDATES:

Candidates should attempt **FOUR** out of the five questions.

All questions carry equal marks. Marks for parts of questions are shown where appropriate.

Data provided:

Filter Tables

Equation list

Four full size Smith Charts – please attach to answer book.

Candidates are not permitted to look at the examination paper until instructed to do so.

1. This question is on link budgets and WIFI.

(a) You are attempting to provide outside WIFI for a friend's garden including to their garage. The relevant information is as follows: The garage is 20 metres away from the house. The cable to the external antenna has a loss of 1dB. The external antenna has a gain of 6dBi. The transmit power in one OFDM subchannel is 10mW and the channel bandwidth is 312kHz. Finally, the noise figure of the WIFI transmitter is 2dB and set for the 5Ghz band ($f=5.8\text{Ghz}$). Calculate, **stating any assumptions**:

- i. The received signal power for a device in the garage.
- ii. The total noise power in the channel and at the receiver.
- iii. The signal to noise ratio.
- iv. Is this above the acceptable 40dB SNR needed for good WIFI and how could it be improved?
- v. In the US the transmit power is allowed to be 100mW. Assuming your answers above are at the limit for operation how much extra range would you get in the US?

[15 marks]

(b) WIFI uses a CSMA-CA method to allow users airtime. Explain with diagrams if needed how this works and what it stands for.

[5 marks]

(c) WIFI is one system that uses MIMO, what is MIMO, what benefits does it provide and how does it achieve the benefits?

[5 marks]

2. This question is about Smith Charts and matching networks.

- (a) You wish to add a passband filter in front of the gain stage you designed in the lab. Unlike the gain stage you used the filter is not matched to 50 ohms. You measure it on the vector network analyser and find its impedance is $10 + 100j$. Plot this on a Smith chart and:
- i. Find the angle and magnitude of the voltage reflection coefficient.
 - ii. The voltage standing wave ratio (VSWR) in linear form and dBs.
 - iii. Plot all values of equal VSWR.
 - iv. What two points on the VSWR plot correspond to the maximum and minimum voltage.
 - v. Find/change the load to an admittance (write your answer as well as marking it on the chart).

[10 marks]

- (b) Design both possible L matches (include your Smith chart) to match the filter chip to 50 ohms and calculate the actual component values required if the centre frequency is 200Mhz.

[15 marks]

3. This question is about stub matching.

- (a) Explain why an impedance of $15+25j$ connected to a piece of transmission line will have different measured values as the distance from it increases and at what distance(s) along the line will it read its original value?

[5 marks]

- (b) Design a single series stub match for the load $15+25j$, include your Smith charts and show all workings. Sketch the layout.

[10 marks]

- (c) For the load $20-150j$ siemens design a parallel double stub match, include your Smith charts, and show all workings. Sketch the layout.

[10 marks]

4. This question is about the radio environment and Channel access.

(a) **Very briefly** explain what each of these stands for/is short for?

- i. FDMA.
- ii. TDMA.
- iii. CDMA-FH and CDMA-SS.
- iv. FDD and TDD.

[4 marks]

(b) What causes fast fading and why is it a problem?

[5 marks]

(c) Why do Doppler effects worsen the problems?

[2 marks]

(d) Give four examples of diversity techniques and explain how they overcome the problem of fading?

[8 marks]

(e) With diverse outputs of 6dB, 8dB and 11dB state and calculate where appropriate the outputs for the diversity combination techniques discussed in the lectures (switched, selection, MG and MR).

[6 marks]

5. This question is about filters.

(a) Using the tables provided design a **passive** low pass Butterworth filter with $F_c = 4$ MHz and $F_s = 8$ MHz to achieve $A_{\min} = 30$ dB and 50Ω scaling resistors.

[10 marks]

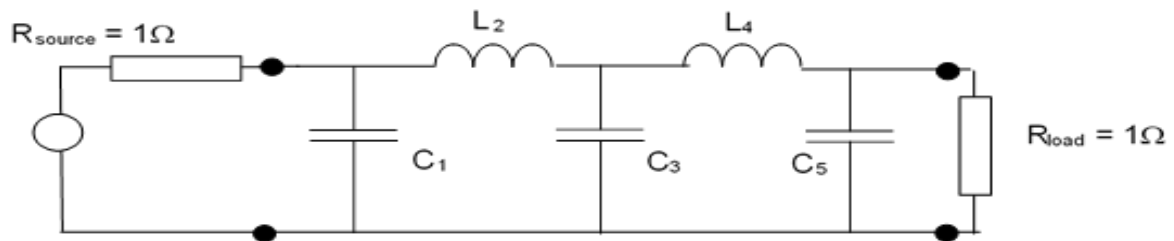
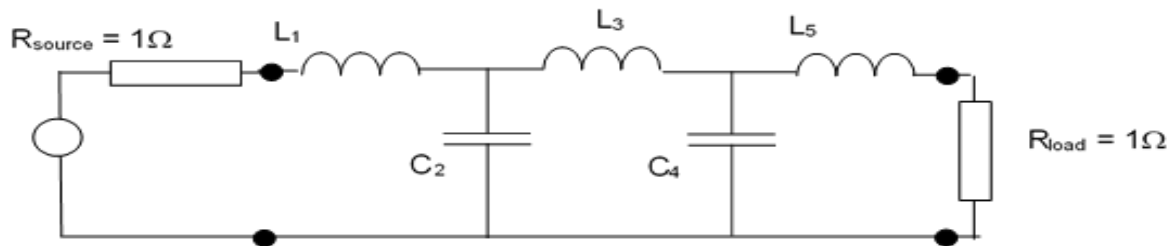
(b) Using the tables provided design an **active** high pass Butterworth filter with $F_s = 2$ MHz and $F_c = 10$ MHz and $A_{\min} = 70$ dB and $10k\Omega$ scaling resistors.

[15 marks]

FILTER DESIGN TABLES

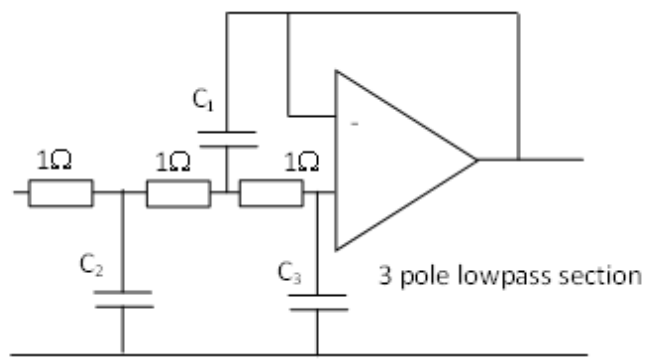
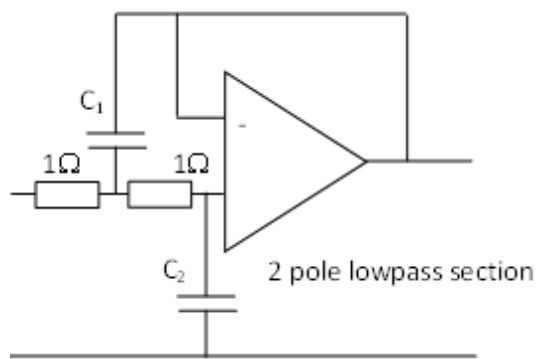
Normalised Butterworth ladder network element values

n	C ₁	L ₂	C ₃	L ₄	C ₅	L ₆	C ₇
2	1.4142	1.4142					
3	1	2	1				
4	0.7654	1.8478	1.8478	0.7654			
5	0.618	1.618	2	1.618	0.618		
6	0.5176	1.4142	1.9319	1.9319	1.4142	0.5176	
7	0.445	1.247	1.8019	2	1.8019	1.247	0.445
	L ₁	C ₂	L ₃	C ₄	L ₅	C ₆	L ₇

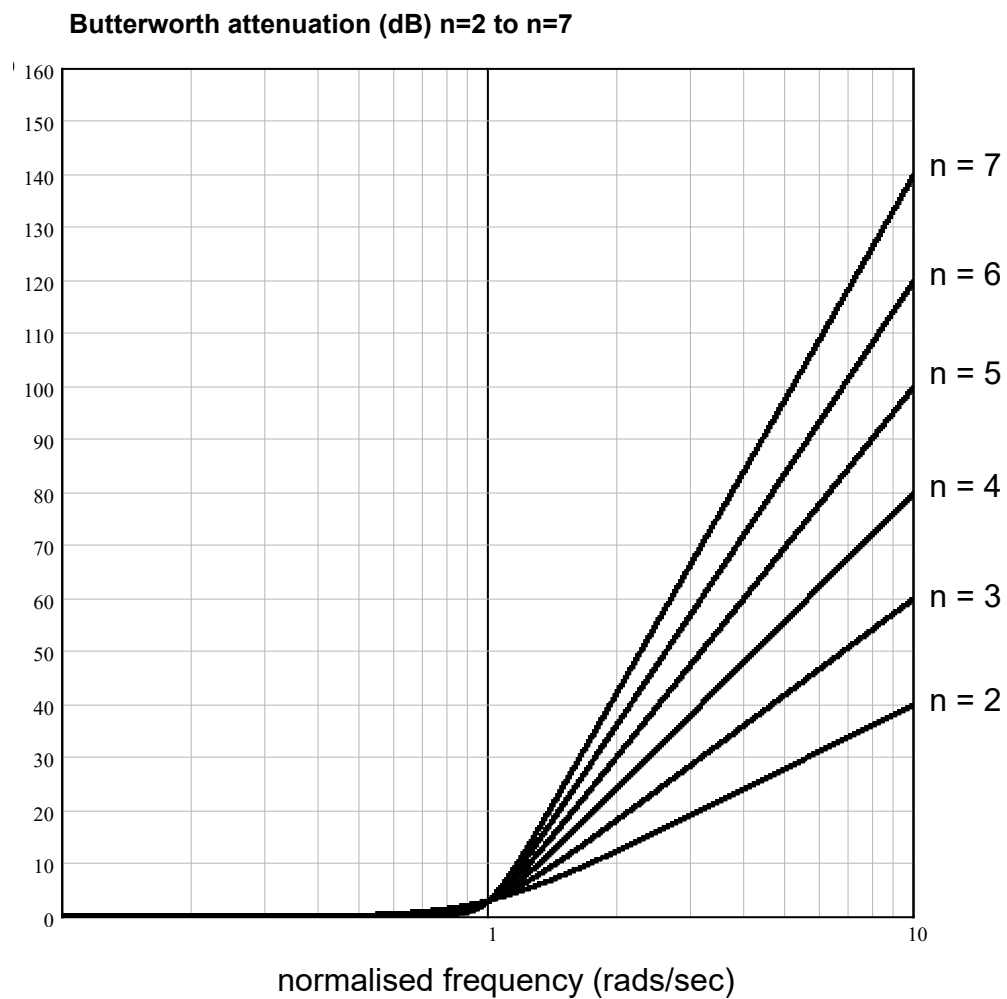
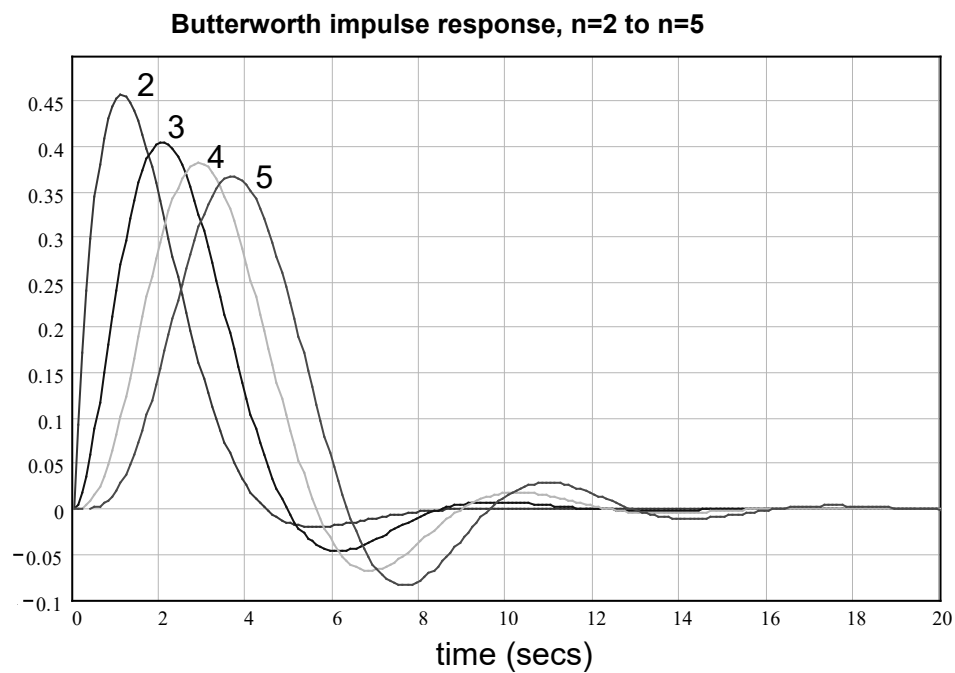


Normalised Butterworth active element values

n	C ₁	C ₂	C ₃
2	1.414	0.7071	
3	3.546	1.392	0.2024
4	1.082 2.613	0.9241 0.309	
5	1.753 3.253	1.354 0.309	0.4214
6	1.035 1.414 3.863	0.966 0.7071 0.2588	
7	1.531 1.604 4.493	1.336 0.6235 0.2225	0.4885



Normalised Butterworth impulse response and attenuation characteristics



Useful Equations

$$L \Rightarrow \frac{LR_o}{2\pi F_c} \quad C \Rightarrow \frac{C}{2\pi F_c R_o} \quad R \Rightarrow R.R_o \quad A_s = \frac{F_s}{F_c}$$

$$R_{HP} = \frac{1}{C_{LP}} \quad \text{and} \quad C_{HP} = \frac{1}{R_{LP}}$$

$$L = 20 \log_{10} \left(\frac{4\pi R}{\lambda} \right) \text{ dB}$$

$$\lambda = V_p / f$$

$$\gamma_{div,2} = \frac{\gamma_1 + \gamma_2 + 2\sqrt{\gamma_1 \gamma_2}}{2} \quad \gamma_{div,N} = \sum_{i=1}^N \gamma_i$$

F = Noise Figure

$$N = -204 + F + 10 \log_{10}(B) \quad (dBW)$$

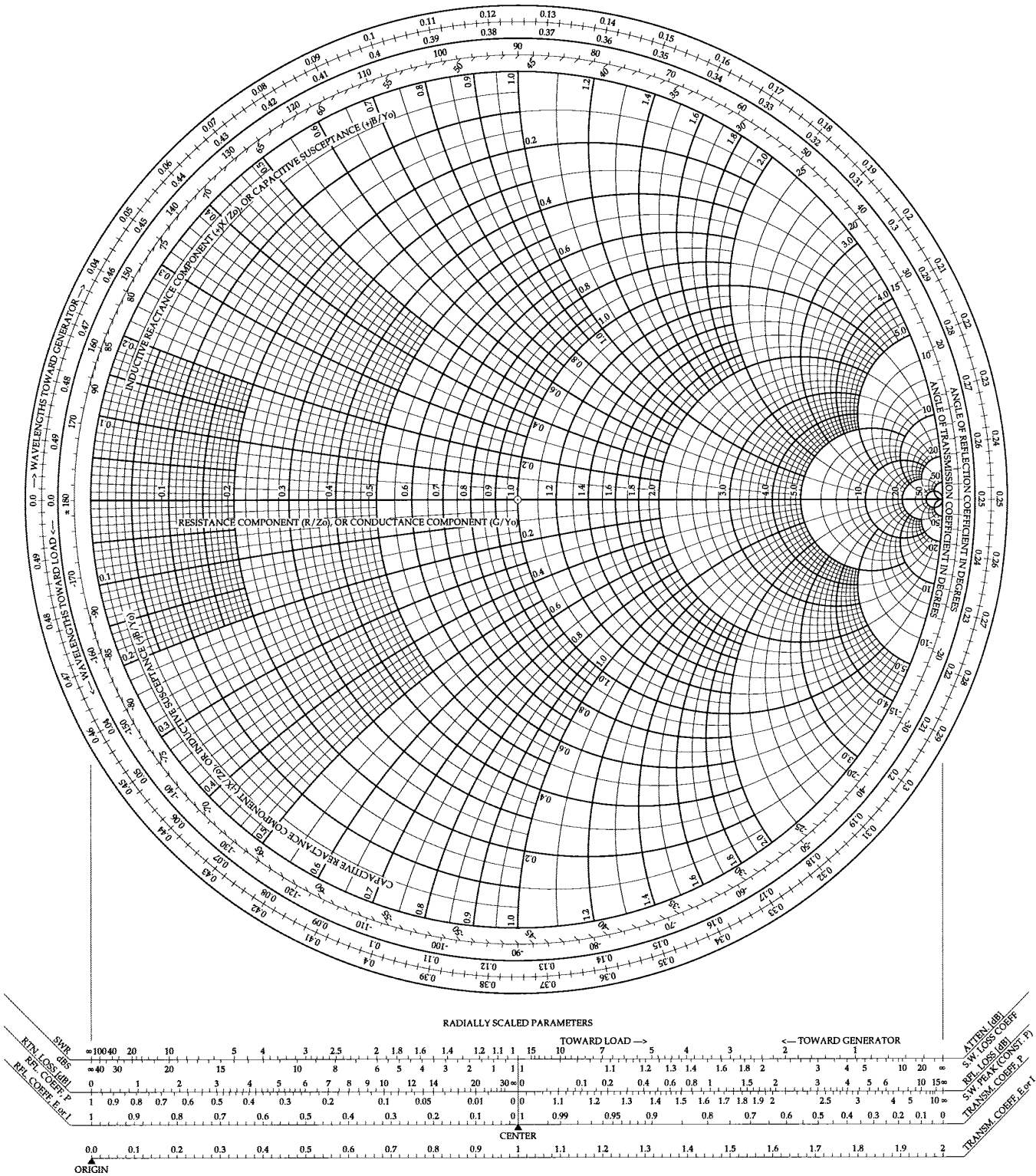
$$N = -174 + F + 10 \log_{10}(B) \text{ dBm}$$

$$\Gamma = \frac{Z - Z_0}{Z + Z_0}$$

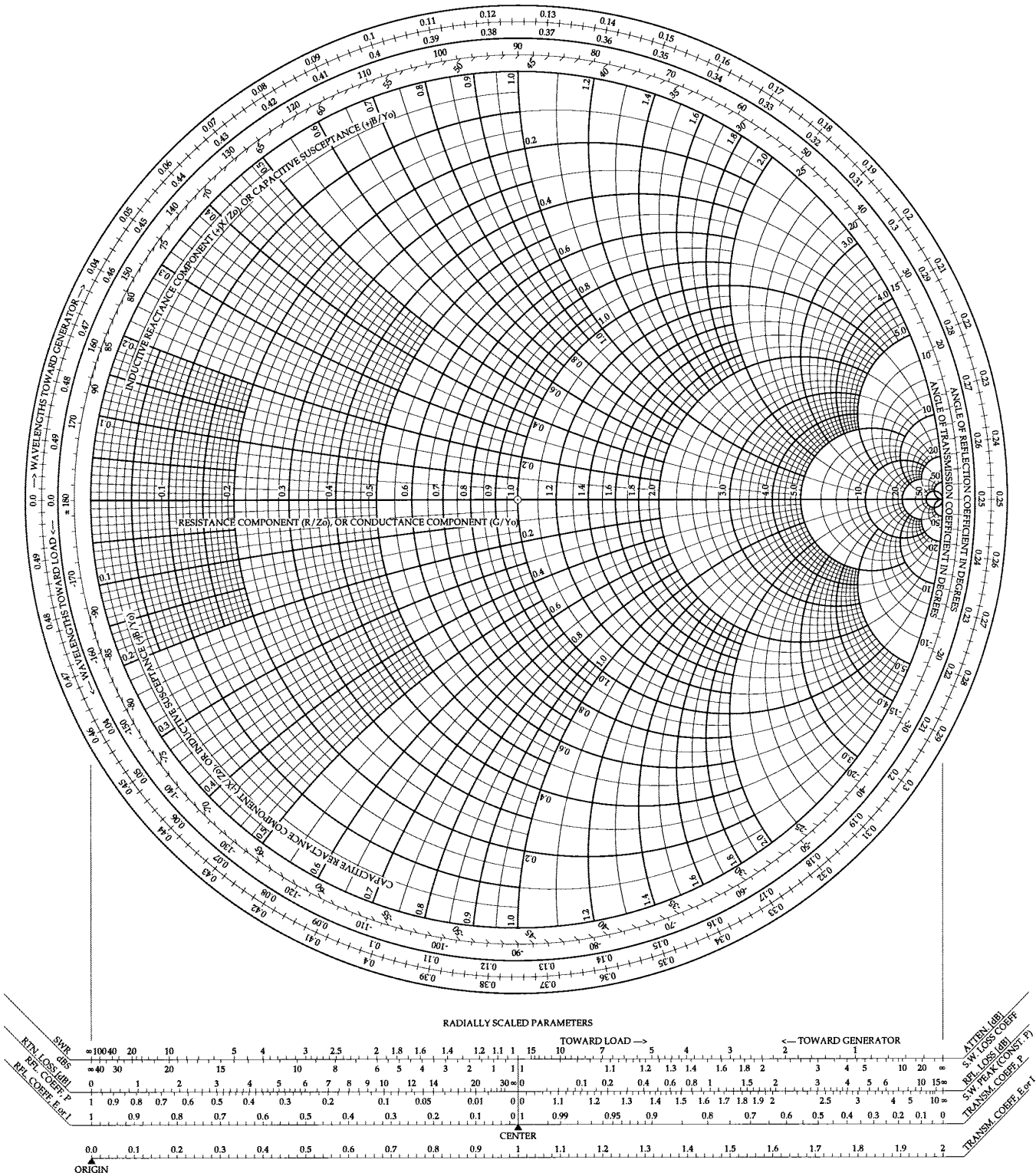
$$\text{return loss} = 10 \log P_r / P_i = -20 \log |\Gamma|$$

$$\text{VSWR} = 1 + |\Gamma| / 1 - |\Gamma|$$

The Complete Smith Chart

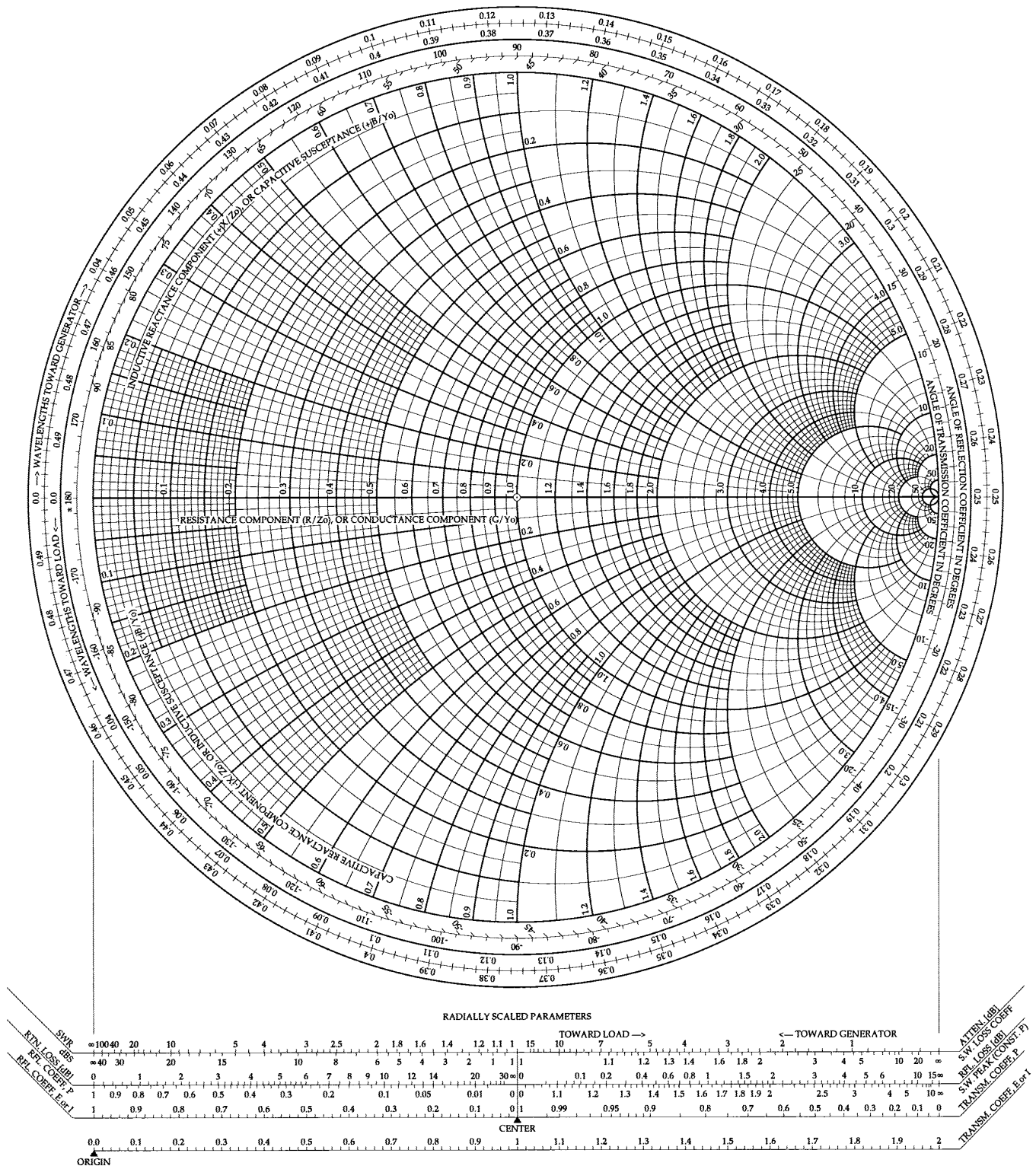


The Complete Smith Chart



Candidate Number Question Number.....

The Complete Smith Chart



The Complete Smith Chart

